Micro-phenomenological Interviews for Hypothesis Generation

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Micro-phenomenological Interviews for Hypothesis Generation

This manual provides instruction for conducting and analysing micro-phenomenological interviews for the purpose of hypothesis generation. It was adapted from the process provided by Valenzuela-Moguillansky & Vásquez-Rosati, 2019, with a view to making it simpler and easier to apply in situations where the goal is solely the generation of hypotheses for later testing. As such, some parts of the original process were removed and other parts were conducted in different ways.

The manual was developed for a pilot study investigating the cognitions, imagination and internal experiences related to engaging with imaginative suggestions, and this example is used throughout. It is divided into sections covering the interview itself, transcript generation and tidying, data preparation, diachronic analysis, synchronic analysis, generic diachronic analysis, generic synchronic analysis, global synchronic analysis, and hypothesis generation.

The Interview

Typically, a number of participants will be interviewed about similar experiences, with Guest, Bunce and Johnson (2006) suggesting between six and twelve to be adequate. Hypothesis generation requires a dependent variable and at least one independent variable. In the example of response to imaginative suggestions, the independent variable was the subjective score given by the participant to their experience (on a Likert scale from 0 representing no effect, to 5 representing the effect felt as real as real) In our example we grouped the scores into three categories: low=0,1; moderate=2,3; high=4,5). The dependent variable was generated from the utterances the participant made during the interview. Essentially, can a participant's description of their experiences, when engaging with a suggestion, be predicted by the subjective score category determined by the participant?

The micro-phenomenological interview itself is a major (and possibly the only) source of data, and because arranging ad hoc second and third interviews can introduce methodological issues (for example, variable time delays between the event and when recalling it), it is strongly recommended to collect all the data that may be required in a single interview per participant. As the investigation intends to generate hypotheses, it should be expected that the researcher does not know, when starting out, what information will become important. The only solution is to ask questions about the entirety of the experience and to be guided by what the participants find important or significant.

In our example, each interview consisted of an event (in our case, an imaginative suggestion during which the participant's eyes were closed) followed by questions about it. The event was clearly time-bounded as starting when the participant closed their eyes, and ending when they were instructed to open them. If your event of interest takes place during the interview, then it should be clearly delineated to focus the participant on the event and not anything that happened before or after it. Use natural markers (such as eyes closure and opening, sounds, specific phrases, etc) to establish the beginning and end of the event.

In our example study, we were specifically interested in cognitions, imagination and internal experiences; however, in order to avoid bias, we started each interview by simply (and patiently) asking the participant to talk us through what just happened. During this part of the interview, ask questions to improve clarity and remove ambiguity. Make notes of anything related to the focus of the study (in our case, the cognitions, imagination and internal experiences related to engaging with the suggestions).

In the analysis phase, we will discern the order of all the experiences they can recall. For each notable item they can recall, therefore, ask if that happened before or after other temporally-related items they have already mentioned. The participant might reveal that the event featured a number of phases; for each, ask about the specific internal experiences that the study is concerned with. In our example, we asked "Did you imagine anything?", "Did you see any pictures?", "What were you aware of?". Note that these questions do not presuppose that the participants imagined or saw anything; in fact, in the pilot study, a number of participants responded that they didn't imagine or see anything for certain phases of the experiences. Had we asked, "What did you imagine/see?" the participants might have felt obliged to report something, even if the real answer was "nothing".

Where answers intersect with the focus of the study (in our case, when a participant mentioned cognitions, imagination or internal experiences) ask for further detail in the direction of the focus of the study. For example, in our study, if a participant's answer mentioned both that their hands felt hot and that they were imagining glue between their fingers, then we would ask for more detail about what was imagined. We would probably not ask for more detail about their hands feeling hot as that is outside of the study's focus, and any more detail is also unlikely to be within the focus.

In our pilot study, we presented three imaginative suggestions individually to each participant. We structured the session with the participant as follows: warm-up exercise, interview about warm-up exercise, suggestion 1, interview about suggestion 1, ditto suggestions 2 and 3. The warm-up exercise and interview did not contribute to the data collection but provided an opportunity for the participant to understand the process and the kinds of questions prior to the real exercises. The warm-up exercise did not involve an imaginative suggestion, but instead a request for a simple, voluntary behaviour. We used a voluntary behaviour because the purpose of the warm-up exercise was to introduce the interview style and was less about the actual event; and also, to prevent any warm-up suggestion having an impact on the responses to the following suggestions.

In summary, ask for an account of what happened, ask questions to determine the order of events, and iteratively ask for more detail in the direction of the study's focus. Give the participant the opportunity to talk for as long as they wish (within the bounds of the experiment).

Transcription

The use of automatic transcription is highly recommended. If sessions are held remotely then the video conference session should be recorded and automatic transcription applied (some video conference solutions include automatic transcription, and standalone automatic transcription services are also available that can work from the recording). In all cases, the transcript needs to be checked and corrected against the recording, utterance by utterance. While time-consuming, this is vastly quicker than manual transcription, and experience to date suggests that every automaticallyproduced transcript contains highly illogical errors that cannot be resolved from the transcripts alone.

The transcript also needs editing for structure and content, and this can be performed at the same time as correcting the transcript, or following as a separate task. In terms of structure, the transcript needs editing so that each line contains only a single utterance, such as a single clause from a sentence, together with the name or identifier of the person speaking. In terms of content, remove all questions where the answers do not rely on the question to make sense. For example, the question, "And what happened next?" can be removed if the answer includes something like, "Well, next I…" as it makes the question superfluous. Superfluous utterances, such as the "Well," in this example can also be removed if the surrounding utterances still project the same meaning with them removed. It is recommended to be thorough when editing for structure and content as effort here will reduce the effort required in the analysis.

Data Preparation

If there were multiple events/interviews within each session (as in our example with three suggestions per session), then the transcripts need splitting into files so that each contains a single interview. The lines of each file then need copying to their own spreadsheet and numbering so that each utterance has its own line number (not spreadsheet cell or row number). In addition to the speaker, utterance and utterance number, the spreadsheets also need the following columns: Moment, IDU (Incipient Diachronic Unit), and Criteria.

The edited transcripts should be preserved, but all analysis should focus on the spreadsheets.

Diachronic Analysis

The purpose of diachronic analysis (within this manual) is to group and order the utterances by moment. Our simplified approach does not include identifying the diachronic structure, and as such does not consider sub-phases of diachronic units.

Following experimentation, it appears that the easiest way to group the utterances by moment is to define categories that the utterances belong to first (specify this in the Criteria column), then reorder the rows based on the Criteria column, and iterate until no further improvements can be made. In this sense, the Criteria should define temporal moments that are distinct from other moments immediately prior or following. Moments can be identified through the participant's use of temporal linkage phrases such as "and then", "after that", "at the beginning", "right at the end", etc. Utterances before a linking phrase should have different criteria to those following it to distinguish the moments. Typical criteria in our pilot study included "The utterances talk about engaging their imagination", and "The utterances talk about feeling their hands move".

After each iteration, the utterances should be reread in the current order (sorted on Criteria) with a view to considering whether the utterances grouped by a single temporal criterion actually represent two or more moments, or whether two or more moments actually appear to happen concurrently and not sequentially. These grouping revisions are made by revising the criteria statements for the relevant utterances, and then the rows resorted, again based on the Criteria column.

Once no further improvements can be found, each criterion can be named in the IDU (Incipient Diachronic Unit) column. Our example criteria could represent IDUs, "Engages Imagination" and "Felt Hands Move". Each IDU can then be numbered in the current utterance order; these should correspond well to experienced moment order, which is likely to be different from the order in which the utterances were originally made. It can be helpful to combine cells within the spreadsheet that all hold the same moment, IDU or criterion.

Synchronic Analysis

Synchronic analysis seeks to discover the themes within each incipient diachronic unit. For each diachronic analysis, create a second spreadsheet work sheet with the following columns: IDU, Utterance number, Utterance, Criteria, ISU, ISU 2nd Level of Abstraction. Copy the IDU, Utterance number, and Utterance for each IDU from the diachronic analysis to the synchronic analysis. While, in theory, the synchronic analysis could be conducted on the original diachronic analysis work sheet, copying the data through and keeping the original preserves the step in the analysis, making it easier to retrospectively appreciate how the data was understood at the end of the diachronic analysis.

Copy the Utterance Numbers, Utterances and IDUs from the diachronic analysis work sheet to the synchronic analysis work sheet. Within each IDU, group utterances by theme, specifying the criteria for the theme in the Criteria column, and the name of the incipient synchronic unit (ISU) the criteria describes. In order to group utterances, it will be necessary to reorder rows within the IDUs. The resultant ISUs within an IDU should, by definition, happen concurrently. If, however, synchronic analysis has revealed themes within an IDU that exhibit temporal order, then this indicates that the IDU represents more than a single IDU and should be split. Return to the diachronic analysis work sheet and apply the change; then copy the changes to the synchronic analysis work sheet.

When the grouping by theme is complete, read through the themes within each IDU to identify if any naturally combine in a higher level of abstraction. If so, name the 2nd level of abstraction and group the ISUs accordingly.

Generic Diachronic Analysis

Once the diachronic and synchronic analyses have been completed for all interviews in the current phase, generic diachronic analysis (and then generic and global synchronic analysis) can take place. Due to the possibility of the synchronic analysis identifying issues with the diachronic analysis, it is not recommended to conduct generic diachronic analysis until the synchronic analysis has been completed.

The purpose of generic diachronic analysis is to identify commonalities between the structure of the IDUs of participants discussing similar events, grouped by the independent variable. Create a new spreadsheet with Participant and Score (or independent variable) as the first two columns. For each participant, enter their name or identifier in the first column, followed by their score (or score category) in the second column; then copy all of the participant's IDUs from the diachronic analysis to the subsequent cells in the same row. Don't try to align IDUs between rows.

Upon completion, there should be a row per participant and reading from left to right should reveal their name/identifier, their score, and all their IDUs in order.

While all participants will have had unique experiences, often the IDUs they report overlap. Group similar IDUs within and between participants by colouring the respective cells with different colours for different groups. In our example, we had a number of IDUs that talked about engaging their imagination, so we coloured them all green; we also had a number of IDUs that talked about feeling bodily movement, so we coloured those yellow. Continuing until either all cells have a group colour or the remaining cells don't belong to any groups, will result in colour-coded, ordered processes.

Through inspection, identify common structures in the different participant rows via the colour coding. Below the participant rows, construct new rows, each containing an identified pattern of coloured generic IDUs. Where similar participant processes exist, group these together by specifying the common elements, and also the optional elements. Ensure each pattern row (through combinations of optional generic IDUs) describes well a set of participant rows. Try to find an optimum set of patterns where all processes can be described by a relatively small set, but where none of the patterns are overly complex to combine notably dissimilar processes. Some trial and error will be required to find the best ways to group the processes. The intention is to identify the differences between the different levels of the independent variable.

Generic Synchronic Analysis

The purpose of the generic synchronic analysis is to bring together the synchronic analyses, group them by event (in our example case, which of the three imaginative suggestions) and by independent variable level or category (in our case, subjective score category: low, moderate or high), so the ISUs can be grouped across participants in the ISU 2nd level of abstraction.

Create a spreadsheet with the same columns as the synchronic analysis, with the addition of a Participant column. Copy this work sheet once for each case (number of events X number of independent score categories). Identify the generic IDUs of interest; these will likely be the generic

IDUs that feature in the summaries of the processes. For each of these generic IDUs, copy the synchronic analyses for the corresponding IDUs they consist of, while labelling the participant column. Inspect the ISUs and any existing 2nd level of abstraction ISUs, and group them within each generic IDU, combining and labelling them in the ISU 2nd Level of Abstraction column. Combine the rows and cells in the work sheet to help show which rows were grouped.

Global Synchronic Analysis

Where the same generic IDUs are found in multiple events (in our example case, where generic IDUs feature in the generic synchronic analyses of multiple suggestions), these generic IDUs can be analysed across the events to reveal even more generic (or global) ISUs or themes, grouped by the independent variable levels.

Create a spreadsheet with the same columns as the generic synchronic analysis, with the addition of an Event (or in our case, Suggestion) column. Copy this work sheet once for each case (number of generic IDUs of interest X number of independent score categories). Copy the generic synchronic analyses to the global synchronic analysis work sheets, while labelling the Event column. Again, inspect the ISUs and any existing 2nd level of abstraction ISUs, and group them within each generic IDU, combining and labelling them in the ISU 2nd Level of Abstraction column. Combine the rows and cells in the work sheet to help show which rows were grouped.

Hypothesis Generation

By developing generic diachronic analyses, generic synchronic analyses, and global synchronic analyses, it is now possible to look for patterns that vary per independent score level, across all participants and events. With a small number of participants, it should be obvious that this analysis only provides weak evidence for these patterns. It is therefore recommended to repeat the process with a second set of participants to see if the same patterns emerge.

To avoid bias, it is recommended to use an independent researcher to conduct the diachronic analysis and synchronic analysis for this phase. After training, an example should be used to compare the analyses of the different researchers, using Cohen's kappa to measure agreement. It is recommended to continue training until $\kappa > .6$.

References

Guest, G., Bunce, A., & Johnson, L. (2006). How Many Interviews Are Enough?: An Experiment with Data Saturation and Variability. *Field Methods, 18*(1), 59-82.

Valenzuela-Moguillansky, C., & Vásquez-Rosati, A. (2019). An analysis procedure for the micro-phenomenological interview. *Constructivist Foundations, 14*(2), 123-145.